# CREDIT SCORE PREDICTION USING MACHINE LEARNING AND DEEP LEARNING

TEAM-3

# TEAM MEMBERS

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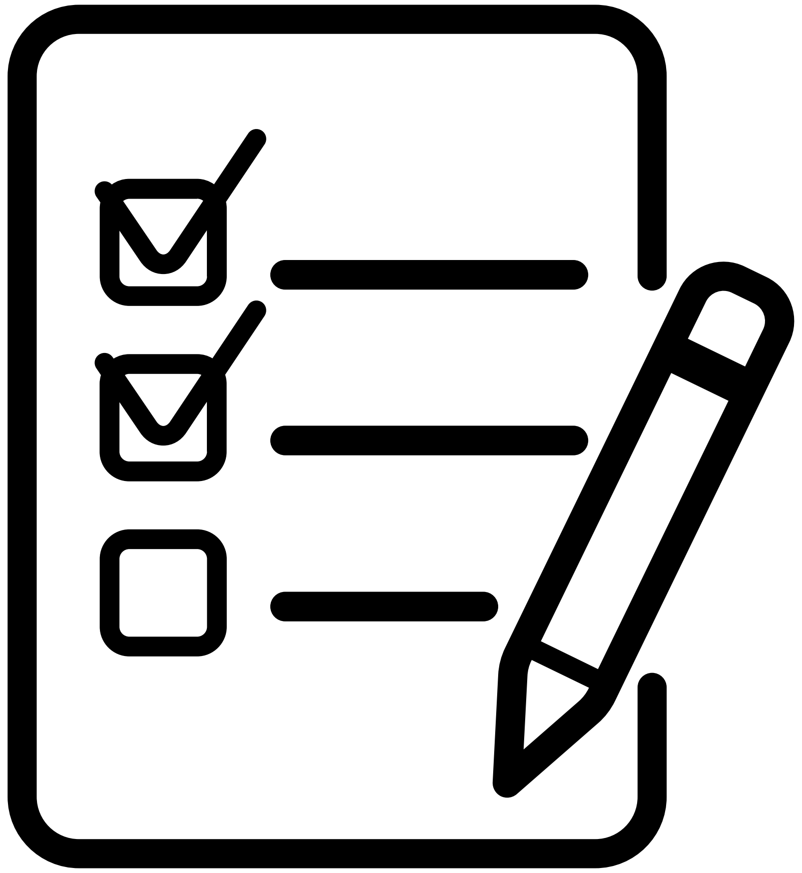
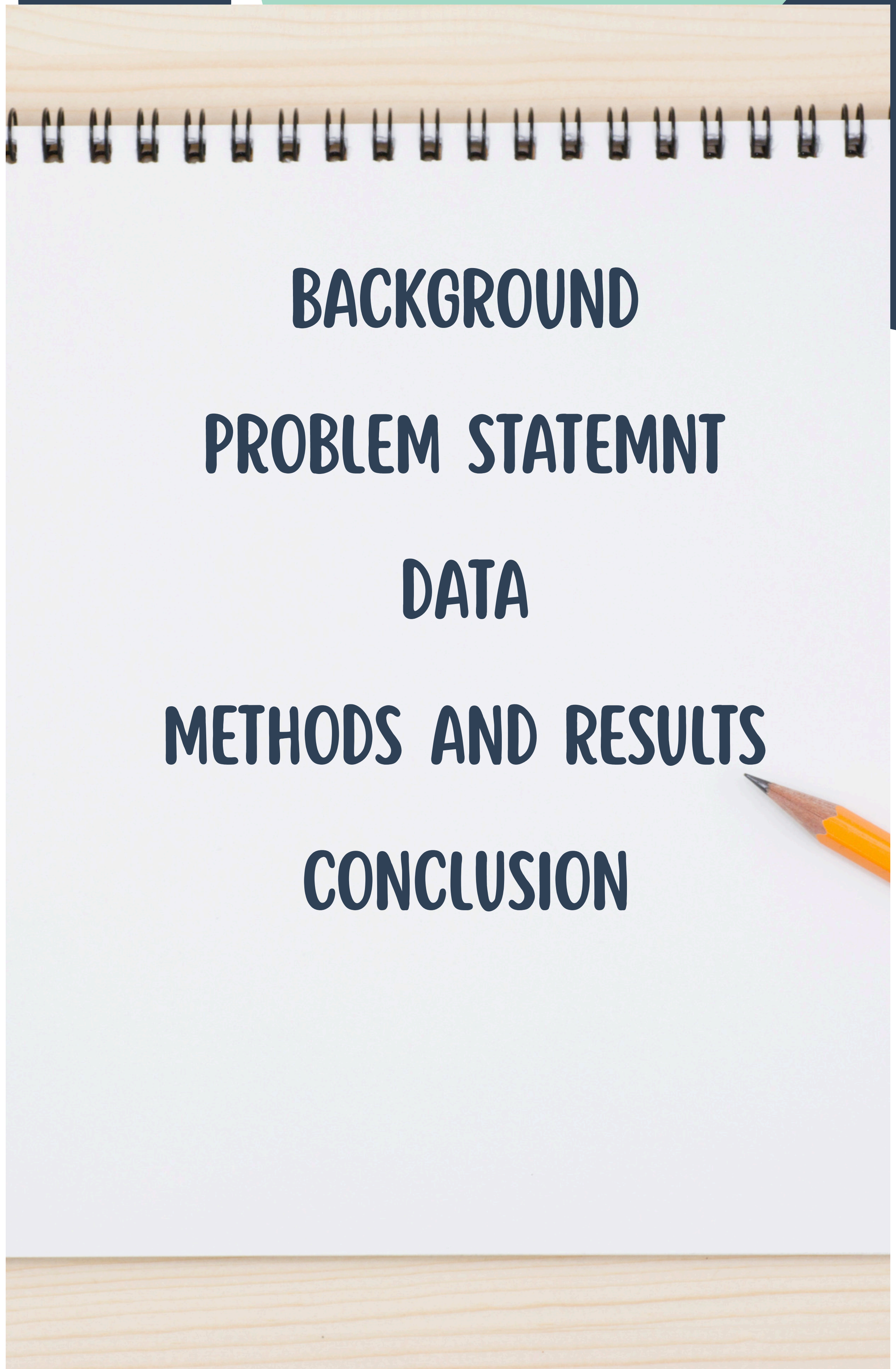
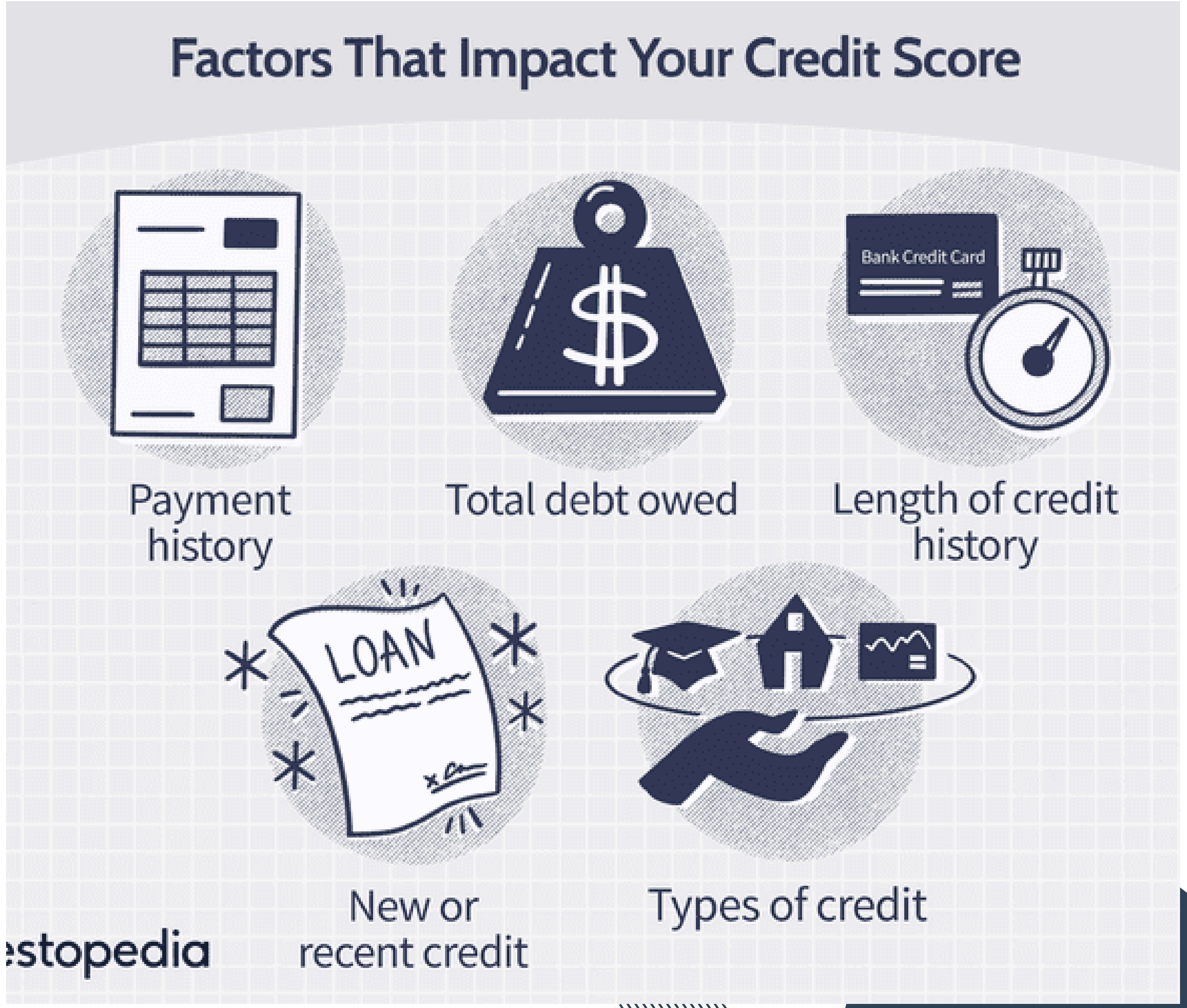
# OUTLINE

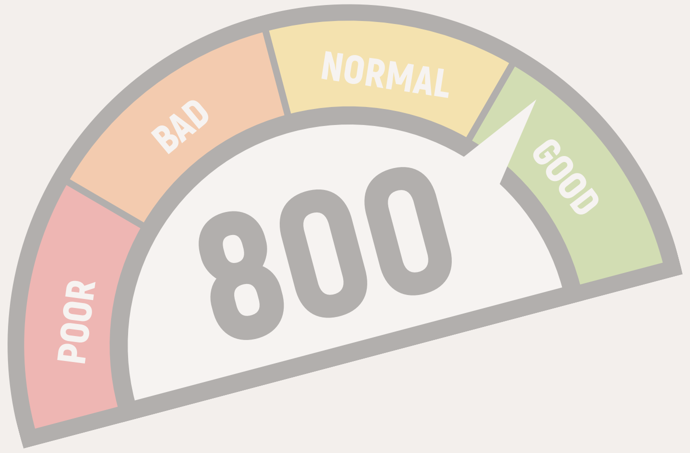



BACKGROUND
PROBLEM STATEMNT
DATA
METHODS AND RESULTS
CONCLUSION

# BACKGROUND

- This project uses machine learning to predict credit scores based on demographic and financial features.
- It enables faster, fairer, and more adaptive credit decisions compared to traditional fixed models like FICO and VantageScore.
- Data-driven techniques allow for personalized credit risk assessment, improving accuracy and inclusivity.






# PROBLEM STATEMENT

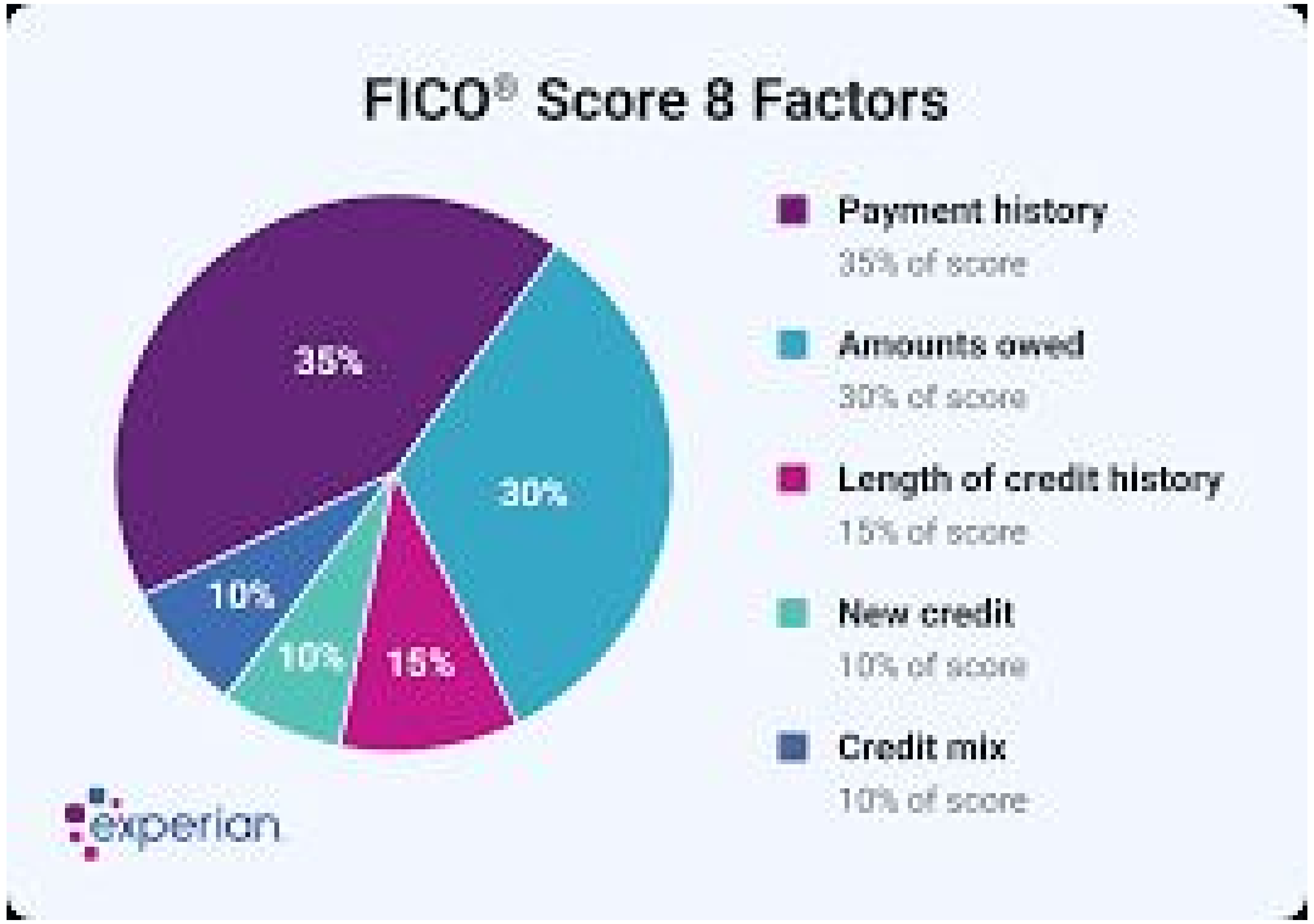
TRADITIONAL CREDIT SCORING MODELS STRUGGLE WITH CAPTURING COMPLEX FINANCIAL BEHAVIORS AND ADAPTING TO EVOLVING TRENDS, LEADING TO INACCURATE CREDIT ASSESSMENTS.

OUR PROJECT LEVERAGES ML AND DL TO IMPROVE PREDICTIVE ACCURACY, FAIRNESS, AND INTERPRETABILITY

# RELATED WORK

TRADITIONAL METHODS: FICO SCORE, VANTAGESCORE.

- EARLY ML APPROACHES: LOGISTIC REGRESSION, DECISION TREES.
- MODERN ML ADVANCEMENTS: RANDOM FORESTS, GRADIENT BOOSTING, NEURAL NETWORKS.
- LITERATURE HIGHLIGHTS THE TRADE-OFF BETWEEN ACCURACY AND INTERPRETABILITY IN CREDIT SCORING.
- OUR STUDY APPLIES MULTIPLE ML MODELS TO IMPROVE PREDICTIVE PERFORMANCE AND OFFER COMPARATIVE INSIGHTS.



STEP 1:

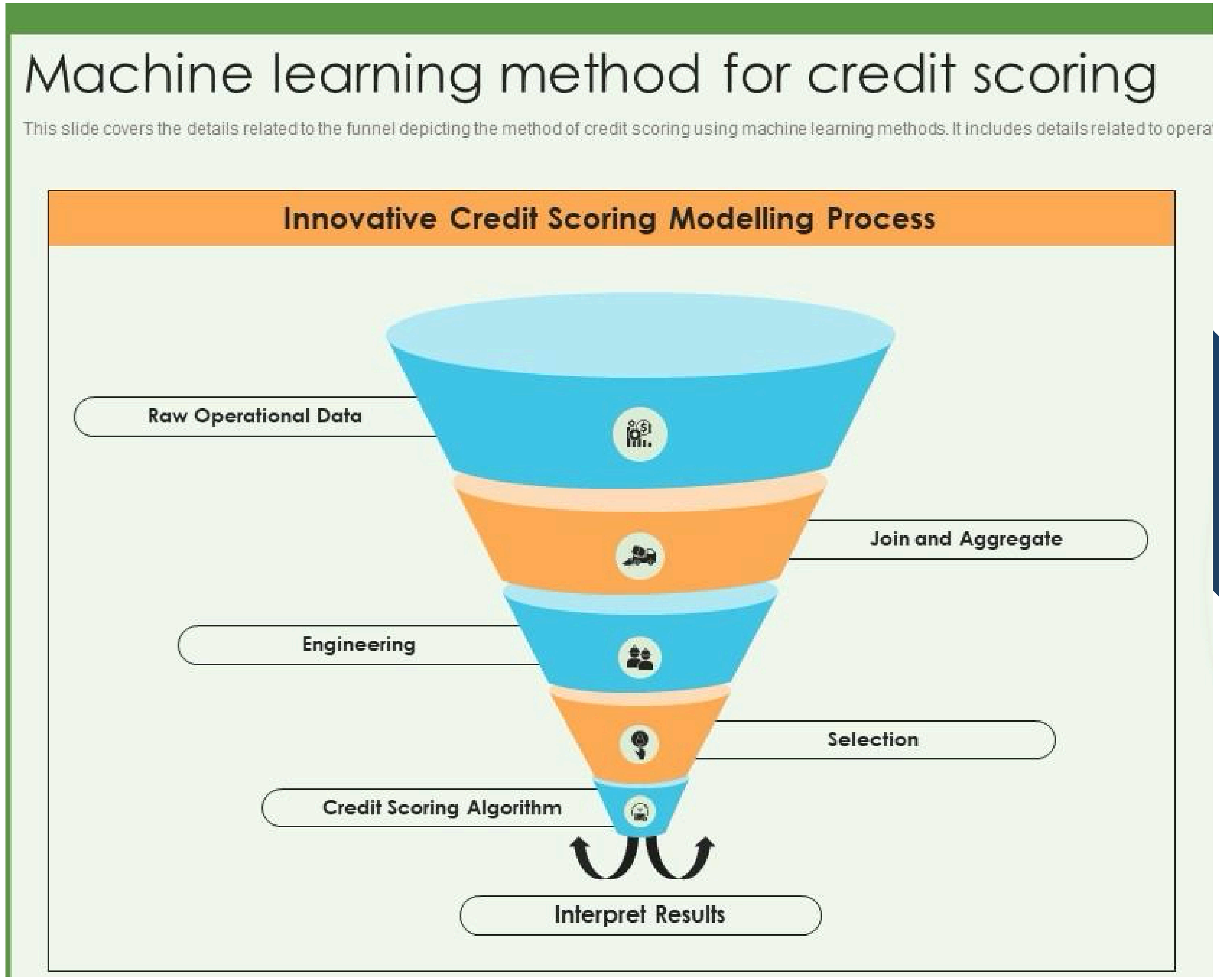
# FURTHER APPROACH

<u>DEVELOPING A MACHINE LEARNING MODEL FOR CREDIT SCORE PREDICTION:</u>

1.BUILD A MACHINE LEARNING MODEL THAT PREDICTS CREDIT SCORES BASED ON VARIOUS FINANCIAL AND BEHAVIORAL FACTORS.

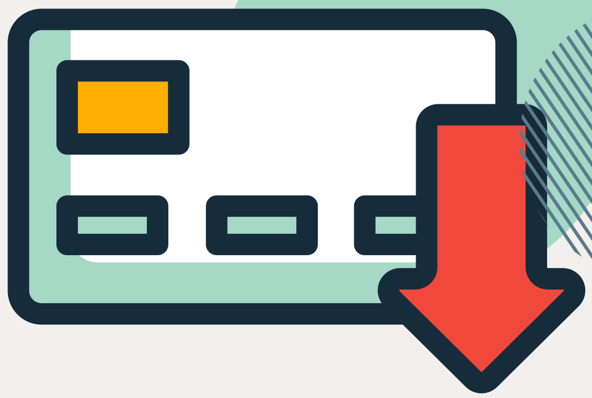
2.EVALUATE DIFFERENT ML ALGORITHMS AND SELECT THE BEST-PERFORMING ONE FOR ACCURATE CREDIT RISK ASSESSMENT.

3.IF THE MODEL'S ACCURACY IS LOW, APPLY HYPERPARAMETER TUNING TECHNIQUES SUCH AS RANDOMIZEDSEARCHCV TO OPTIMIZE PERFORMANCE AND IMPROVE PREDICTIVE ACCURACY.



# FURTHER APPROACH

## STEP 2:

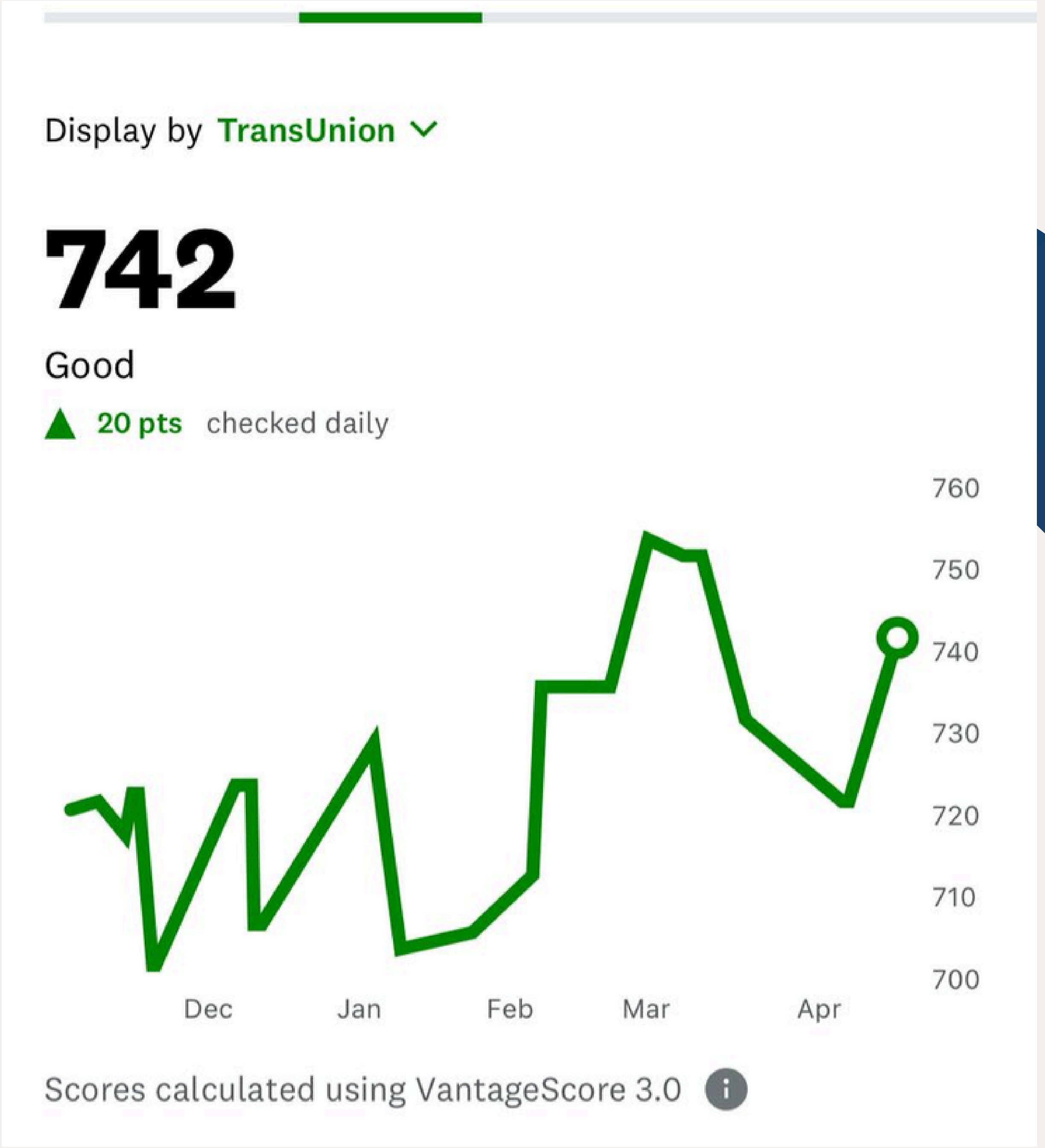


**MODEL DEPLOYMENT AND WEB-BASED APPLICATION DEVELOPMENT:**

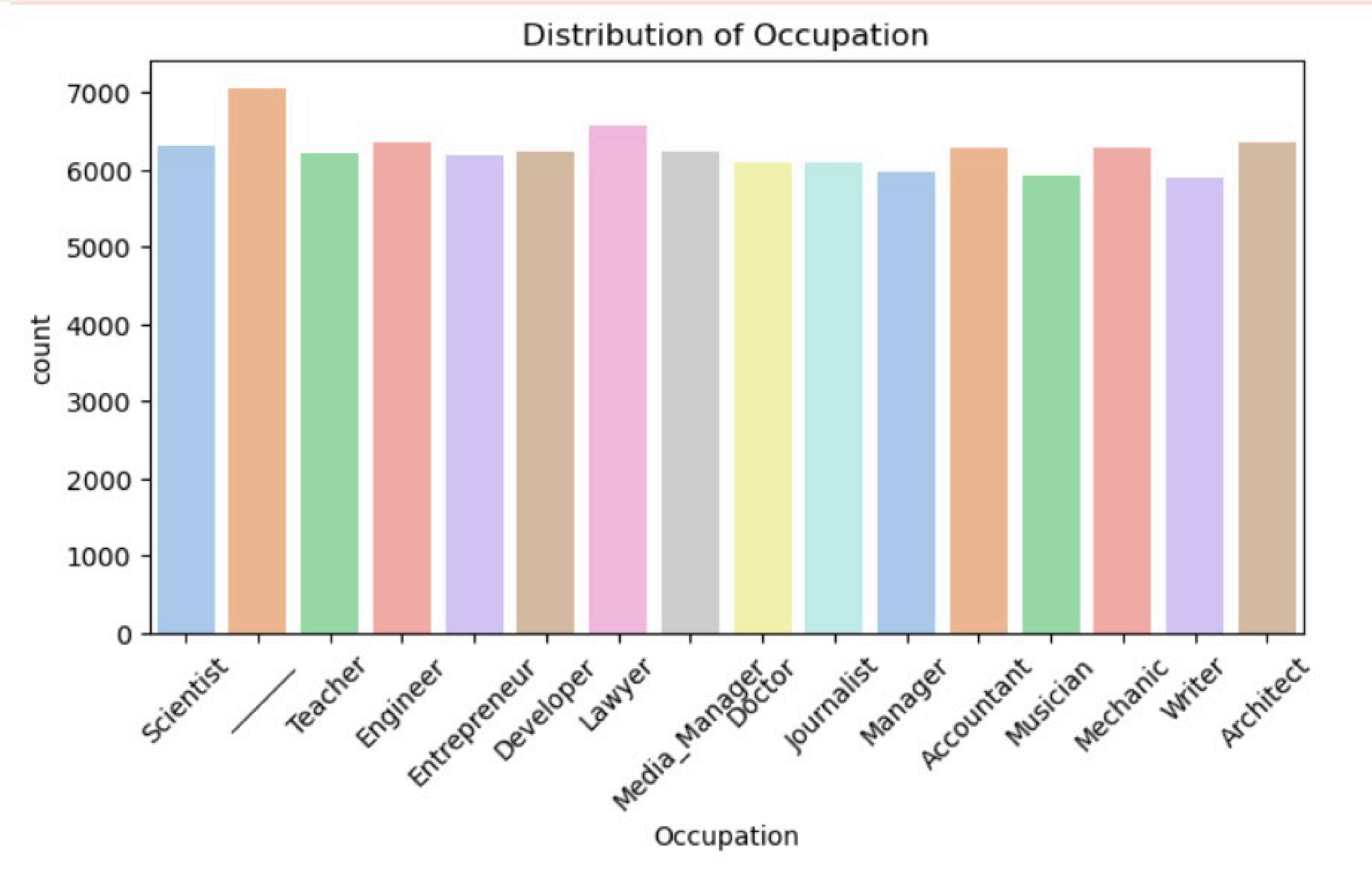
1.ONCE THE OPTIMIZED MODEL IS FINALIZED, SAVE IT AS A PICKLE (.PKL) FILE FOR FUTURE USE.

2.DEVELOP A WEB-BASED APPLICATION THAT INTEGRATES THE TRAINED MODEL, ALLOWING USERS TO INPUT FINANCIAL DETAILS AND RECEIVE PREDICTED CREDIT SCORES.

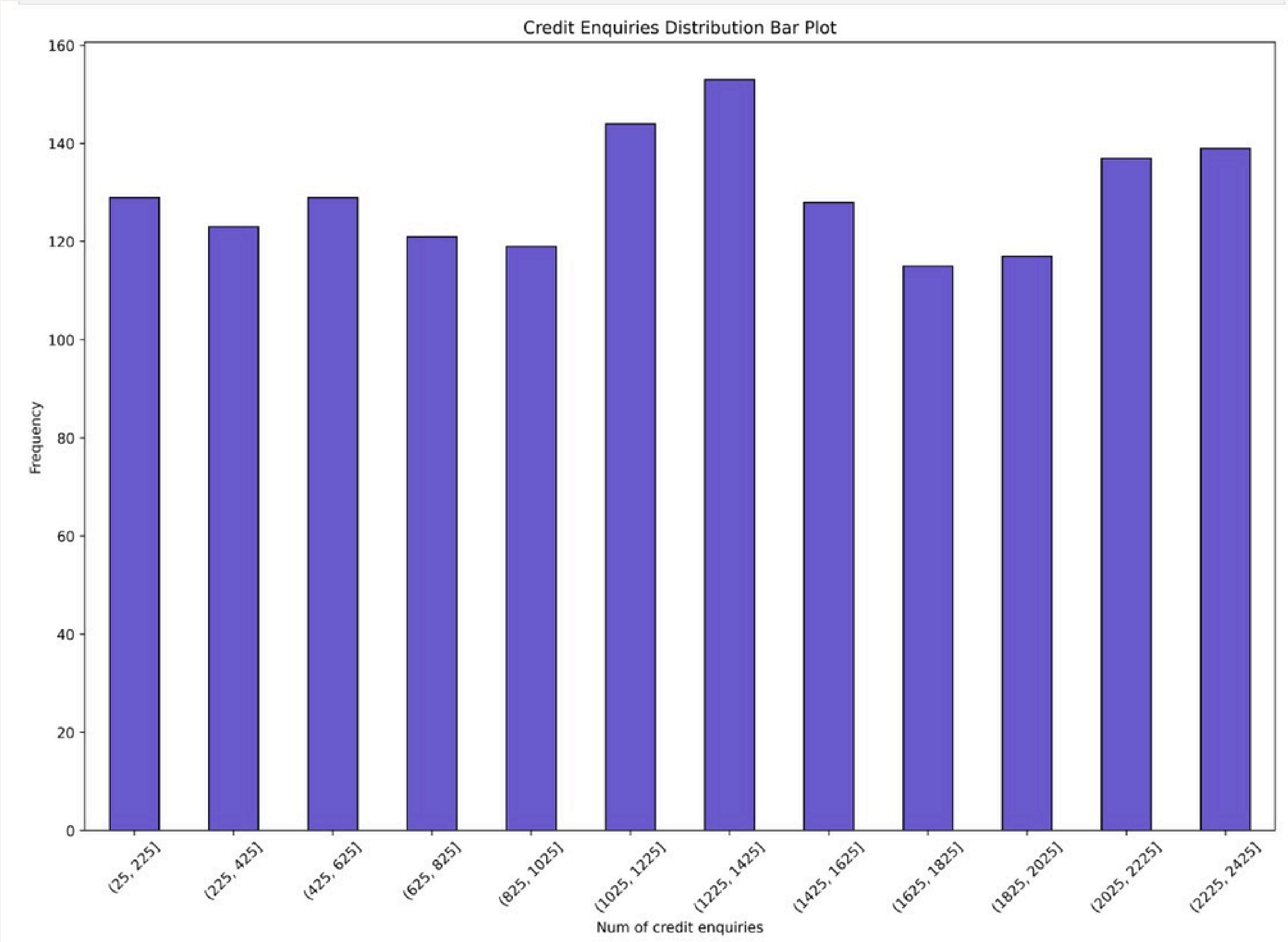
ENSURE THE WEB APP IS USER-FRIENDLY, INTERACTIVE, AND PROVIDES CLEAR EXPLANATIONS OF PREDICTIONS USING XAI TECHNIQUES (E.G., FLASK) FOR TRANSPARENCY



# EXPLORATORY DATA ANALYSIS (EDA)

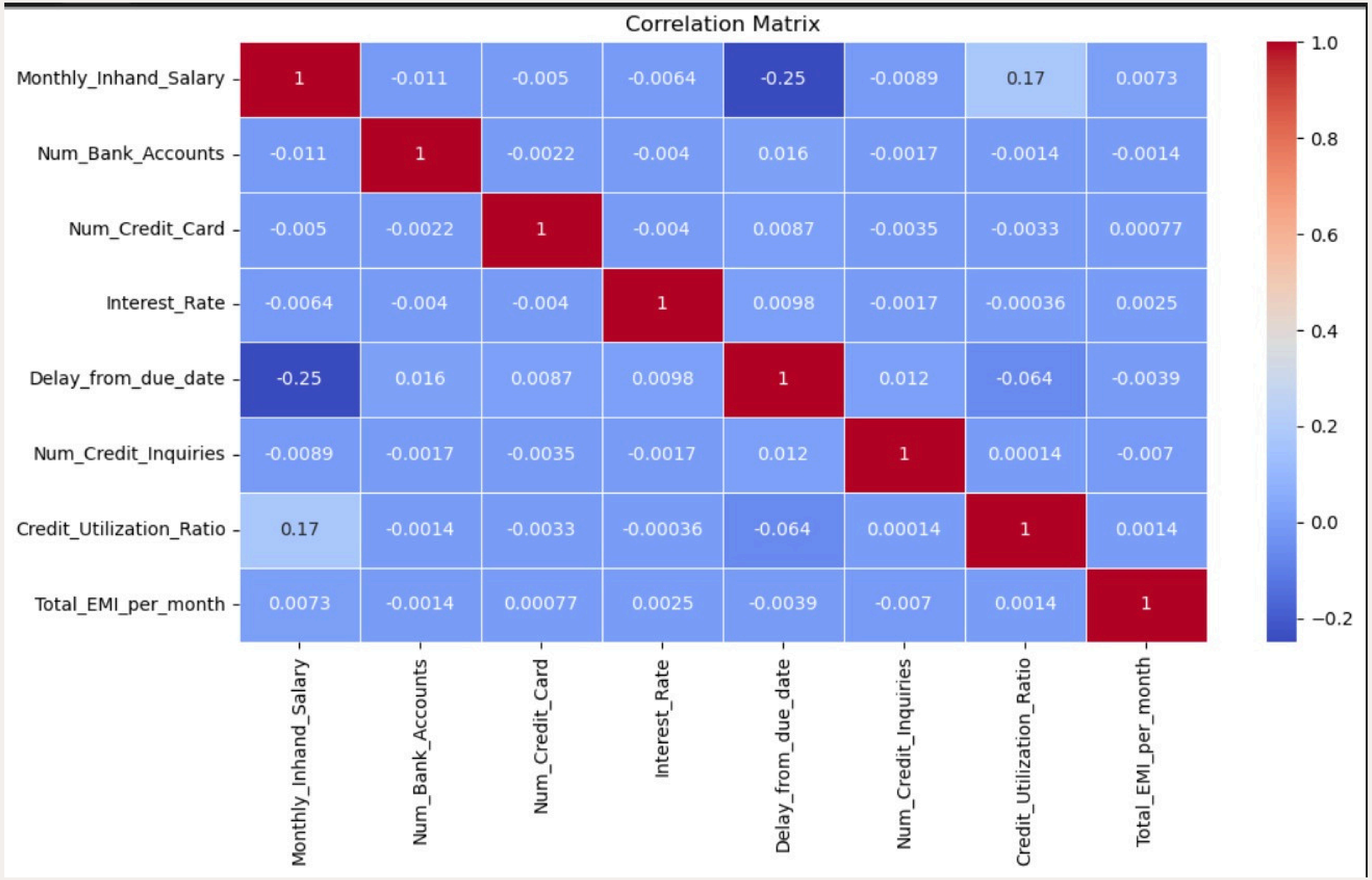



REPRESENTATION OF DISTRIBUTION OF OCCUPATION VS COUNT IN THE DATASET

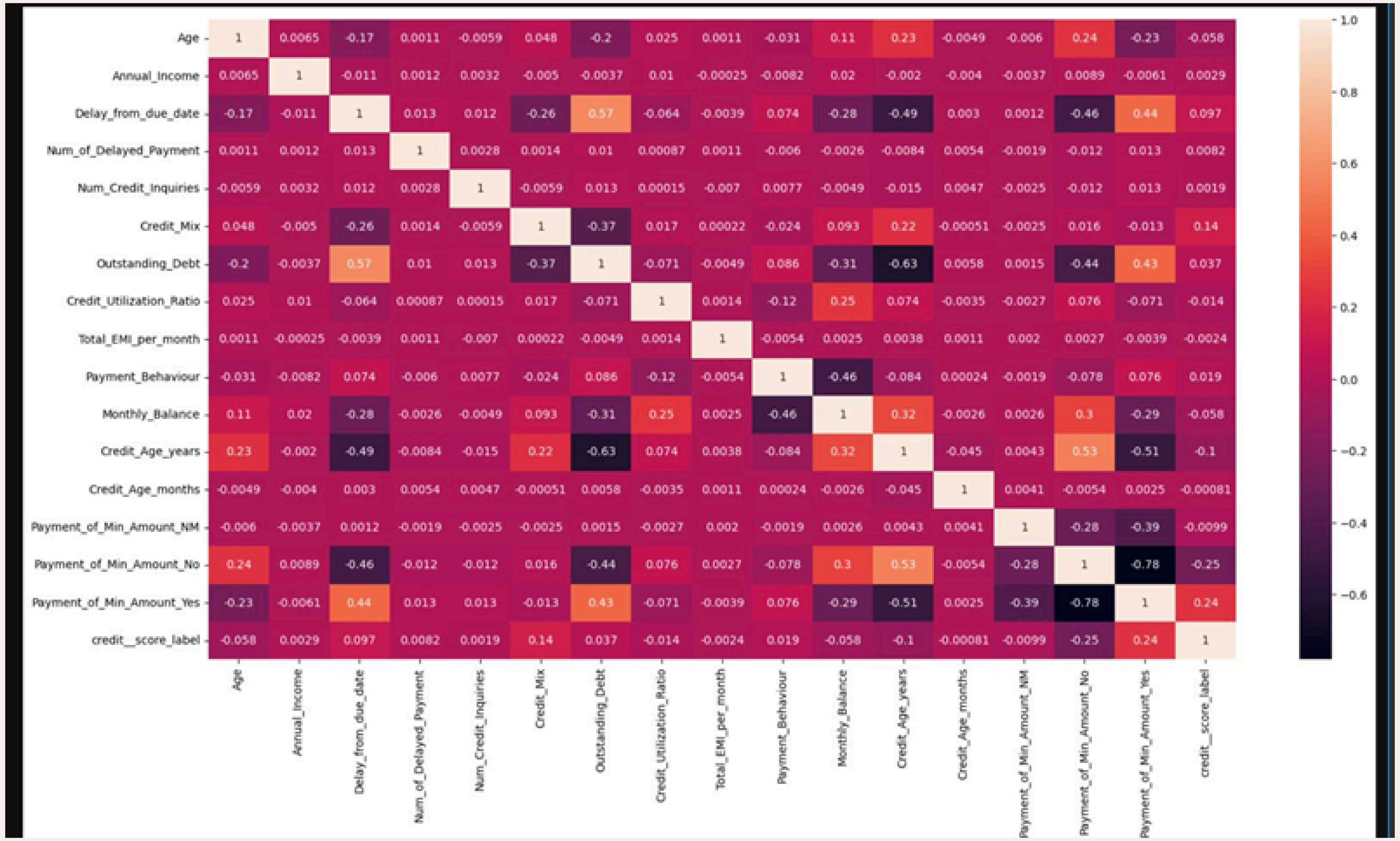



REPRESENTATION OF CREDIT ENQUIRIES DISTRIBUTION BAR PLOT

# CORRELATION MATRIX BEFORE DATA CLEANING



# CORELATION MATRIX AFTER DATA CLEANING




OUTSTANDING DEBT AND DELAYED PAYMENTS SHOW A MODERATE POSITIVE CORRELATION WHILE CREDIT AGE AND THE NUMBER OF DELAYED PAYMENTS ARE NEGATIVELY CORRELATED.

FACTORS AFFECTING CREDIT SCORE

# CHI- SQUARE TEST RESULT

CHI-SQUARE STATISTIC: MEASURES DEVIATION FROM EXPECTED FREQUENCIES; HIGHER VALUES INDICATE STRONGER RELATIONSHIPS.

- P-VALUE: PROBABILITY UNDER THE NULL HYPOTHESIS (NO RELATIONSHIP).
- $P < 0.05$ → SIGNIFICANT ASSOCIATION
- $P > 0.05$ → NO SIGNIFICANT ASSOCIATION

DEGREES OF FREEDOM (DF): NUMBER OF INDEPENDENT COMPARISONS.

KEY FINDINGS:

- AGE ($\chi^2$ = 5229.79, P = 0.0) → STRONG IMPACT ON CREDITWORTHINESS.
- OCCUPATION ($\chi^2$ = 181.1, P = 2.43E-24) → MODERATE INFLUENCE.
- ANNUAL INCOME ($\chi^2$ = 134413.15, P = 0.0) → MAJOR FACTOR IN CREDIT SCORING.
- DELAY FROM DUE DATE ($\chi^2$ = 24626.21, P = 0.0) → SIGNIFICANT NEGATIVE IMPACT.
- NUMBER OF DELAYED PAYMENTS ($\chi^2$ = 20540.30, P = 0.0) → FREQUENT DELAYS LOWER CREDIT SCORES.

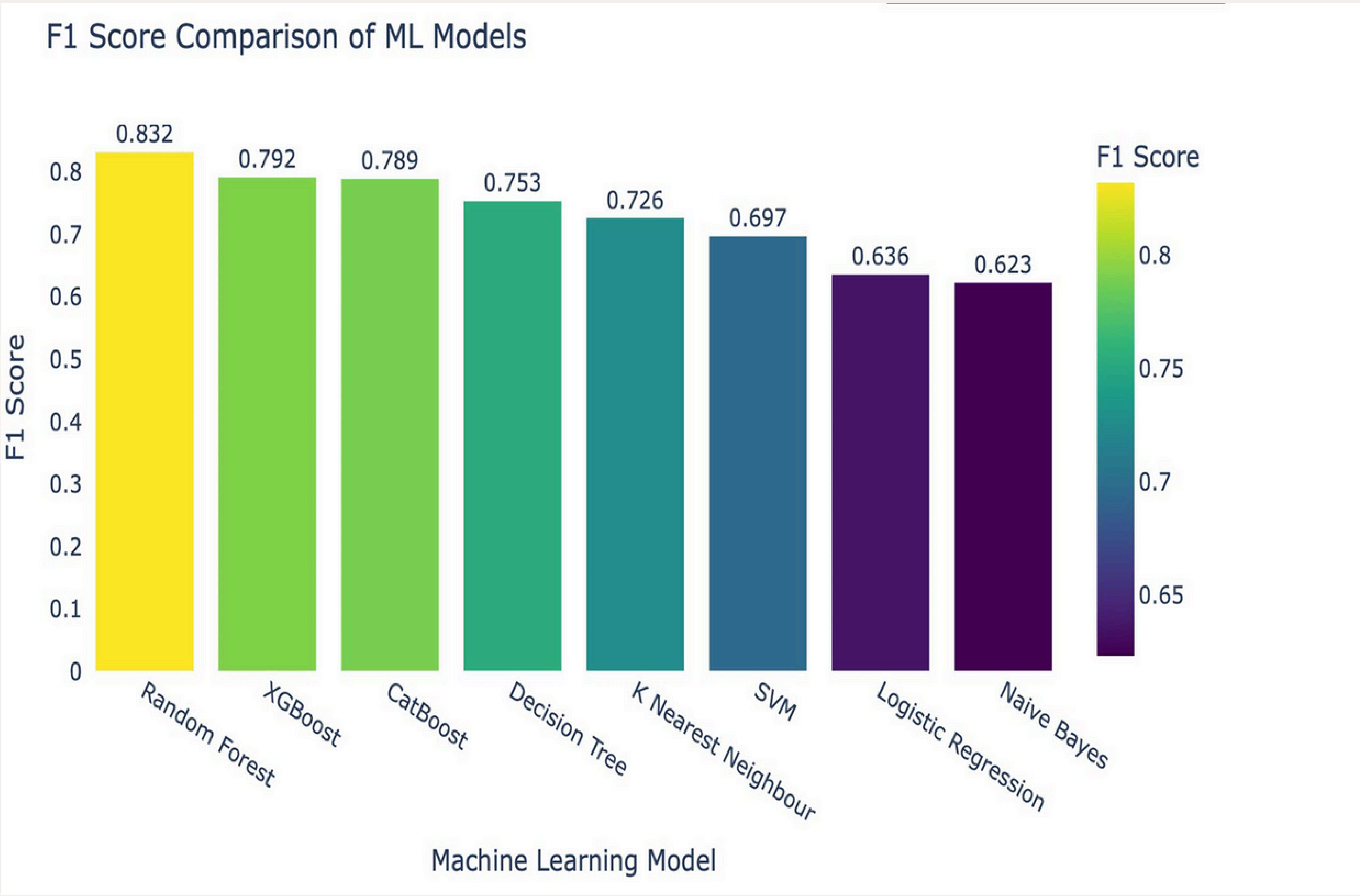
CONCLUSION

- ALL VARIABLES SIGNIFICANTLY IMPACT CREDIT SCORES.
- ANNUAL INCOME, DELAYED PAYMENTS, AND AGE ARE THE STRONGEST PREDICTORS.

# MACHINE LEARNING MODELS

- LOGISTIC REGRESSION - LINEAR MODEL PREDICTING CLASS PROBABILITIES
- K-NEAREST NEIGHBOUR - INSTANCE-BASED LEARNING USING FEATURE SIMILARITY
- NAIVE BAYES - PROBABILISTIC MODEL ASSUMING FEATURE INDEPENDENCE
- DECISION TREE - RULE-BASED, INTERPRETABLE MODEL
- RANDOM FOREST - ENSEMBLE OF DECISION TREES (BAGGING)
- SVM - MARGIN-BASED CLASSIFIER EFFECTIVE IN HIGH DIMENSIONS
- XGBOOST - FAST, REGULARIZED GRADIENT BOOSTING
- CATBOOST - BOOSTING OPTIMIZED FOR CATEGORICAL FEATURES

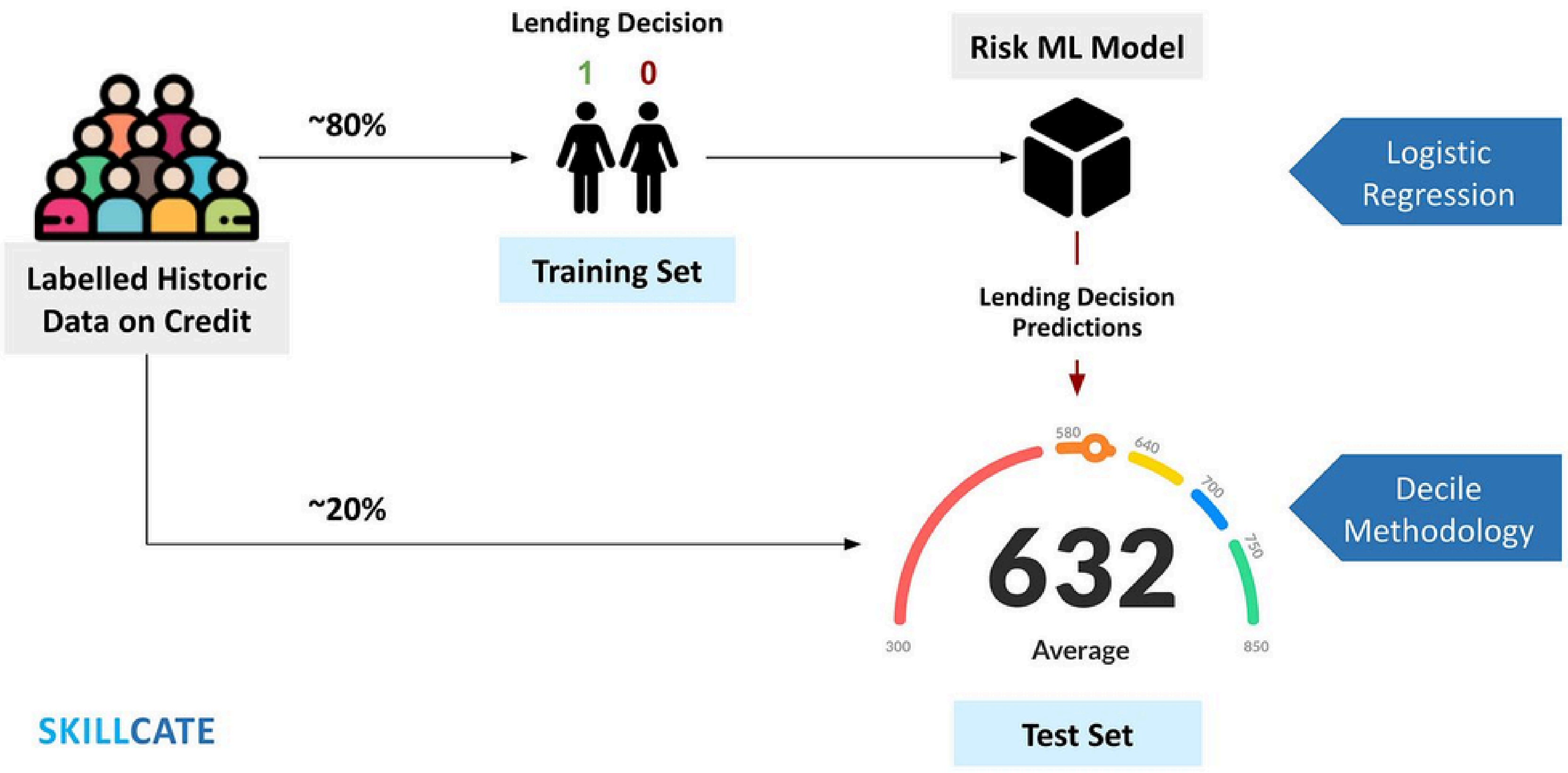
# MACHINE LEARNING MODEL RESULTS:



# DEEP LEARNING MODELS:

## FEEDFORWARD NEURAL NETWORK - MULTILAYER DENSE ARCHITECTURE

- LAYERS: 1286432 → SOFTMAX OUTPUT
- ACTIVATION: RELU + DROPOUT + BATCHNORMALIZATION
- OUTPUT ACTIVATION: SOFTMAX FOR MULTICLAS
- FRAMEWORK: TENSORFLOW WITH KERAS API



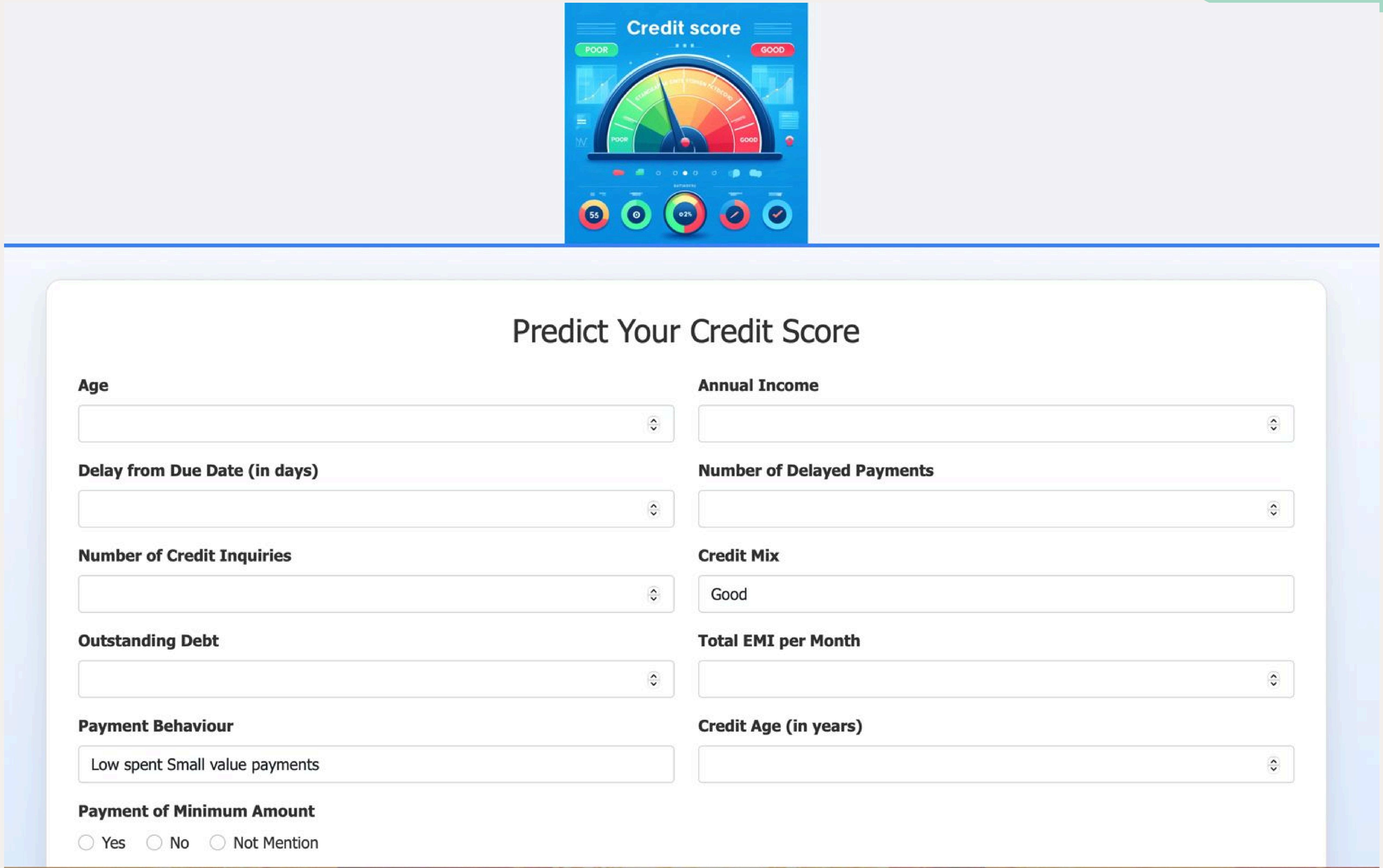

- FEATURES:

CLASS WEIGHT BALANCING
EARLY STOPPING
FEATURE SCALING WITH STANDARDSCALER

# LINK TO WEBSITE

MASON.GMU.EDU/~CKURAKUL

"WE BUILT A USER-FRIENDLY WEB APPLICATION USING FLASK, HTML, AND CSS THAT PREDICTS CREDIT SCORE CATEGORIES BASED ON USER INPUTS SUCH AS INCOME, DEBT, PAYMENT BEHAVIOR, AND CREDIT HISTORY."



# RESOURCE PAGE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5. RIBEIRO, M. T., SINGH, S., & GUESTRIN, C. (2016). “WHY SHOULD I TRUST YOU?” EXPLAINING THE PREDICTIONS OF ANY CLASSIFIER. IN PROCEEDINGS OF THE 22ND ACM SIGKDD INTERNATIONAL CONFERENCE ON KNOWLEDGE DISCOVERY AND DATA MINING (PP. 1135– 1144).
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# THANK YOU